For a **20 kg turbojet-powered UAV** (typically utilizing a 85N to 100N thrust turbine like a JetCat P100 or KingTech K85), the fuel system requirements change dramatically compared to a gasoline piston engine. Turbojets require **high-pressure delivery, kerosene-compatible materials, precise electronic metering, and absolute immunity to air ingestion** to prevent catastrophic flameouts.

The industry-standard, highly reliable fuel system configuration for a 20 kg jet UAV consists of the following architecture:

1. Fuel Storage and Kerosene Compatibility

- **Main Tank:** Typically a custom-molded Kevlar/fiberglass composite tank or a flexible bladder tank contoured to the fuselage saddle over the Center of Gravity (CG).
- **Critical Material Warning:** Unlike gasoline setups, **silicone tubing and standard rubber will degrade and dissolve when exposed to Jet-A, Jet-A1, or Kerosene.** You must use **Tygon F-4040-A tubing, Viton lines, or specialized polyurethane (PU) tubing** with aluminum or brass fittings and O-rings.

2. The Hopper Tank (Air Trap / Header Tank)

In a high-speed turbojet UAV executing rapid pitches and high-G turns, fuel sloshes violently. An air bubble entering a turbine fuel line causes an immediate flameout and turbine shutdown.

- Install a **250ml to 500ml rigid hopper tank** (often called a UAT - Ultimate Air Trap or bubble trap) between the main tank and the fuel pump.
- Inside the hopper tank, a weighted **felt clunk or ceramic filter clunk** remains submerged in fuel regardless of aircraft orientation, ensuring that only bubble-free fuel is pulled into the pump.

3. High-Pressure Fuel Pump and ECU Integration

Unlike piston engines that use low-pressure diaphragm carburetors, turbojets operate under high pressure:

- **Electric Fuel Pump:** A brushless or brushed high-pressure gear pump (controlled directly by the turbine's **ECU - Electronic Control Unit**). It must be capable of delivering fuel at pressures ranging from **3 to 6+ bar (45 to 90+ psi)** at maximum throttle.

- **Solenoid Valves:** Integrated electronic valves managed by the ECU handle automatic startup (opening propane/gas or controlling direct kerosene ignition lines) and act as an instant safety fuel shutoff in the event of an emergency or aborted takeoff.

4. Filtration and Fuel Management

- **Dual-Stage Filtering:**
  A. A high-flow filter at the fuel filling station (fuel dot).
  B. A fine-mesh micro-filter positioned *between* the main tank/hopper and the fuel pump to protect the precision gears of the high-pressure pump from microscopic debris.
- **Fuel Oil Mixture:** Remember that jet fuel (Jet-A/Kerosene) lacks lubricating properties. A **2% to 5% mixture of high-grade 2-stroke turbine oil** (such as Aeroshell Turbine Oil or specialized synthetic turbine oil) must be premixed into the fuel to lubricate the fuel pump bearings and turbine internal bearings (unless using independent lubrication systems).

Summary Checklist for a 20kg Turbojet Fuel Setup

Component	Specification	Purpose
Main Tank	Kevlar / Composite / Bladder	Centrally mounted over the CG for stable weight distribution.
UAT / Header Tank	250ml–500ml Bubble Trap	Prevents air bubbles from entering

		the fuel line during high-G flight.
Tubing	Tygon F-4040 or Viton	Withstands hydrocarbon breakdown from Jet-A/Kerosene.
Pump & ECU	High-pressure gear pump + FADEC/ECU	Delivers 3–6 bar pressure mapped precisely to throttle commands.
Clunk Filter	Felt or Ceramic Clunk	Filters out fine particulate and traps final traces of air.

TTo calculate the required fuel mass, tank volume, and fuel consumption rate for a 20 kg turbojet UAV, you need to work backward from the engine's thrust requirement and its

Thrust Specific Fuel Consumption (TSFC).

Step 1: Determine Thrust Required for Cruise

For a 20 kg aircraft in steady, level cruise, the engine does not need to produce 20 kg of thrust (approx. 196 N). Instead, thrust must equal aerodynamic drag:

$$\text{Thrust Required } (T) = \frac{\text{Total Weight}}{\text{Lift-to-Drag Ratio } (L/D)}$$

- **Typical Turbojet UAV L/D:** ~6 to 10.
- **Calculation (assuming $L/D = 8$):**

  $$T = \frac{20\text{ kg}}{8} = 2.5\text{ kg of thrust} \approx 24.5\text{ N}$$

Accounting for climb, maneuvering, and acceleration margins, average cruise thrust typically operates at roughly **30 N to 40 N** (~3 to 4 kgf).

Step 2: Calculate Mass Fuel Consumption Rate

Small micro-turbojets (e.g., JetCat P100, KingTech K85) have a TSFC of roughly **0.12 to 0.15 kg / (N · hr)** at cruise throttle settings.

Using a conservative TSFC of **$0.14\text{ kg/(N}\cdot\text{hr)}$** at an average cruise thrust of **35 N**:

$$\text{Fuel Flow Rate } (\dot{m}) = T \times \text{TSFC}$$

$$\dot{m} = 35\text{ N} \times 0.14\text{ kg/(N}\cdot\text{hr)} = 4.9\text{ kg/hr}$$

Converting this to a per-minute rate:

$$\text{Per-minute consumption} = \frac{4.9\text{ kg/hr}}{60} \approx \mathbf{0.0817\text{ kg/min}} \quad (\approx 81.7\text{ g/min})$$

Step 3: Calculate Required Usable Fuel Mass

For a 15-minute primary flight mission:

$$\text{Mission Fuel Mass} = 15\text{ min} \times 0.0817\text{ kg/min} = \mathbf{1.225\text{ kg}}$$

Adding a standard **20% fuel reserve** (for taxi, climb, wind headroom, and landing setup):

$$\text{Total Fuel Mass Required} = 1.225\text{ kg} \times 1.20 = \mathbf{1.47\text{ kg}}$$

Step 4: Convert Mass to Tank Volume

Jet-A / Kerosene has a density of approximately **0.80 kg/L** at standard temperature (15°C).

$$\text{Total Tank Volume Required} = \frac{\text{Total Fuel Mass}}{\text{Fuel Density}}$$

$$\text{Total Tank Volume Required} = \frac{1.47\text{ kg}}{0.80\text{ kg/L}} = \mathbf{1.8375\text{ Liters}} \approx \mathbf{1,840\text{ ml}}$$

Summary Specification Matrix

Parameter	Value	Notes
Average Cruise Thrust	~35 N (3.57 kgf)	Based on $L/D = 8$ with flight margins
Fuel Flow Rate	**~102 ml/min** (81.7 g/min)	At cruise throttle
15-Min Mission Fuel	1.225 kg (~1.53 L)	Pure mission burn
Safety Reserve (20%)	0.245 kg (~0.31 L)	Covers taxi, takeoff, and landing
Total Tank Capacity	**~1.85 Liters** (1,850 ml)	Split between Main + Hopper Tank

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System Sizing Recommendation

To achieve this 1850 ml total capacity safely on your 20 kg frame:

1. **Main Tank: Install a 1,500 ml composite saddle tank located directly over the Center of Gravity.**

2. **Hopper Tank (UAT): Install a 350 ml Ultimate Air Trap tank downstream of the main tank to feed bubble-free fuel directly to the high-pressure pump**

   Because turbojet UAVs operate at significantly higher speeds than propeller-driven drones, parasite drag (skin friction and form drag) plays a much larger role than induced drag
3. **.To maximize the Lift-to-Drag (L/D) ratio and fuel efficiency of a 20 kg turbojet UAV, you must optimize three primary areas: airfoil selection (wing profile), wing planform geometry, and cruise speed matching.**

1. Wing Profile (Airfoil) Selection & Optimization

1. **Unlike high-lift, highly-cambered airfoils used on slow surveillance drones, a turbojet UAV requires a low-drag, medium-thin profile designed for higher Reynolds numbers.**
   - **Thickness-to-Chord Ratio (t/c): Aim for a thickness of 10% to 12%. Thinner sections reduce form drag and delay compressibility effects at high speeds, while still leaving enough internal height for a structural spar.**
   - **Camber Selection: Choose a low-to-moderate camber. High camber creates excessive pitching moments and drag at high speeds. Profiles like the MH-32, E1210, or specialized laminar-flow supersonic/transonic sections help maintain extended laminar boundary layers, reducing skin friction drag.**
   - **Surface Finish: Maintain an extremely smooth, glossy finish on the composite wing skin. Surface roughness trips the boundary layer into turbulence prematurely, sharply increasing skin friction drag.**

2. Planform and Geometry Optimization

1. **How the wing is shaped in plan view dictates how efficiently it generates lift relative to its span.**
    - **Aspect Ratio (AR): Induced drag is inversely proportional to aspect ratio ($AR = b^2/S$). While ultra-high aspect ratios (like gliders) cause structural weight penalties on a heavy jet, maintaining an AR between 7 and 9 provides an optimal sweet spot between low induced drag and structural rigidity.**
    - **Taper Ratio and Twist (Washout): Use a taper ratio of roughly 0.5 to 0.6 (tip chord shorter than root chord) to approximate an elliptical lift distribution, which naturally minimizes induced drag. Add 1 to 2 degrees of geometric washout (twist) at the wingtips to prevent tip stall during high-G maneuvers.**
    - **Winglets / Tip Fences: Adding clean, blended winglets or raked wingtips reduces wingtip vortex drag, effectively increasing the aerodynamic aspect ratio without adding massive wingspan.**

3. **Cruise Speed and Operating Point Tuning**
    1. **For a turbojet, fuel consumption is directly proportional to thrust produced. To maximize efficiency, you must fly at the exact airspeed where aerodynamic efficiency (L/D) is maximized.**
        - **Fly at Maximum L/D Speed (V_{md}): Maximum range for a jet aircraft occurs precisely at the angle of attack and airspeed that yields the maximum L/D ratio (where parasite drag equals induced drag). You can calculate this target cruise velocity using:**
          $$V_{cruise} = \sqrt{\frac{2 \cdot W}{\rho \cdot S \cdot C_{L\text{_at_max_L/D}}}}$$
          (Where W is aircraft weight, ρ is air density, S is

wing area, and C_L is the lift coefficient corresponding to the peak of the L/D curve).

- **Avoid High-Speed Drag Spikes: Turbojet fuel burn skyrockets if pushed past its optimal cruise envelope into transonic drag divergence zones. Keep cruise speeds safely below the critical Mach number (M_{crit}) of your chosen airfoil.**